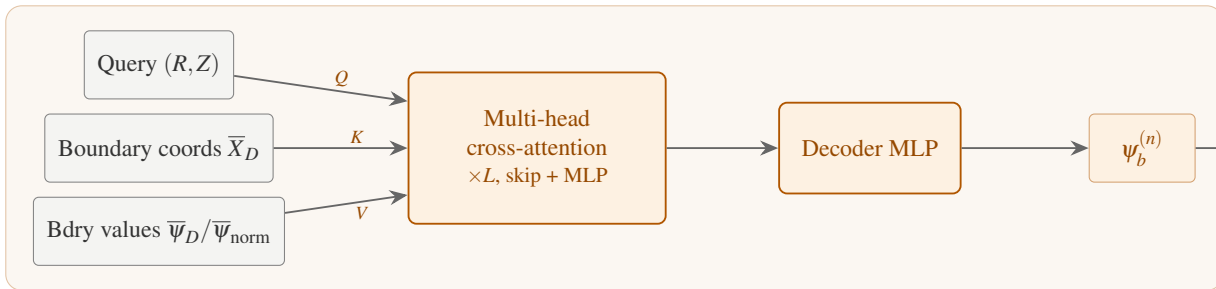
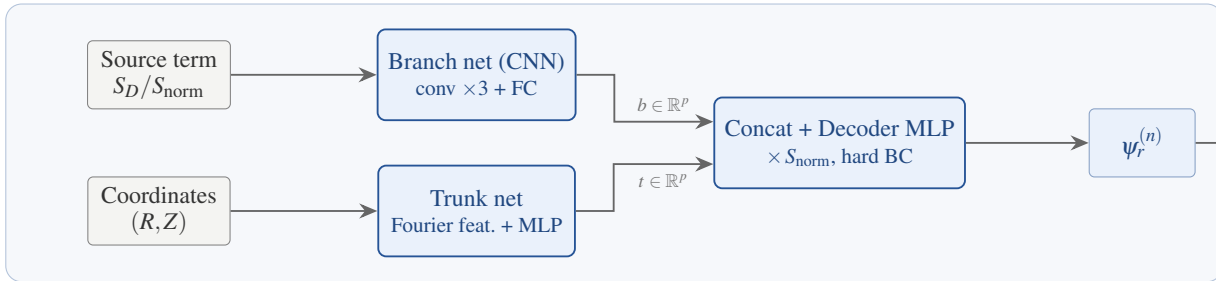


PI-DeepONet (residual part $\mathcal{N}_r$)



PINTO (boundary part $\mathcal{N}_b$)

$$\psi^{(n)} = \psi_r^{(n)} + \psi_b^{(n)}$$

Both sub-models are trained separately with a physics-informed loss, without solution labels.